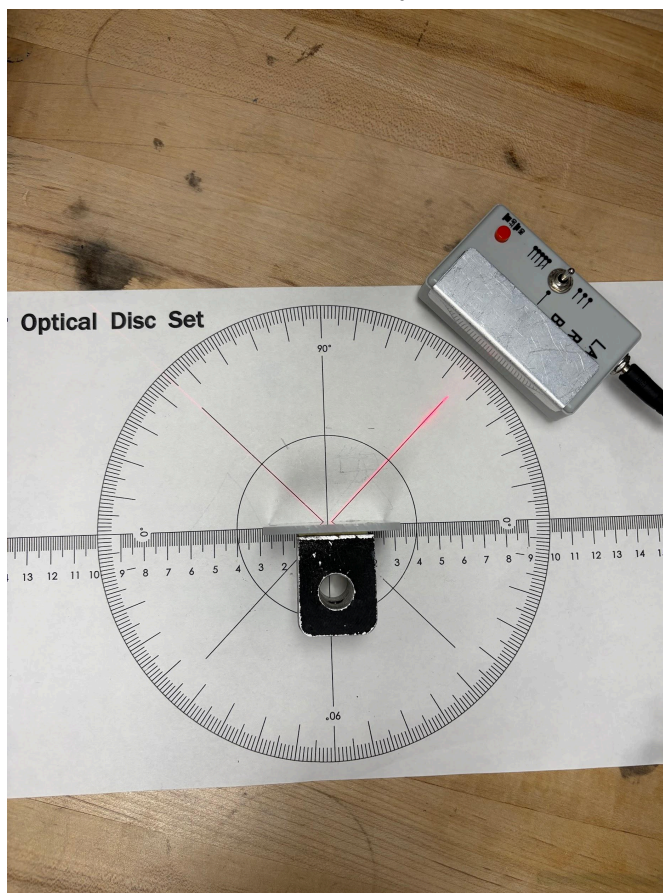


Exercise - Parallax

- **Observation (pencils are not equidistant from the eye):** Both of the pencils wobble; the pencil closer to the eye appears to wobble more.
- **Observation (pencils are equidistant from the eye):**
- **Experimental Design:** We can determine the distance between the observer and a nearby object by measuring the angular displacement of the target object as compared to the angular displacement of a separate object of known distance.

Experiment 1 - The Law of Reflection

- **Calibration:** Using the clearly marked 45 degree lines, we shine the laser light towards the mirror at 45 degrees. Move the mirror up or down along the plane of reflection until the reflected beam aligns with the other 45 degree reference line. At this position, the reflection surface will lie directly on the line of the protractor.



Difficulties

- The mirror has a finite thickness, so is difficult to align with the 0 point of the protractor. It would be far simpler if it were infinitely thin.
- The laser beam is also not infinitely thin, so it is difficult to determine its exact centre. We account for this somewhat by taking the uncertainty of our measurements to be the thickness of the beam.

Verify/Refine:

Θ_i (°)	Θ_{r_1} (°)	Θ_{r_2} (°)	Θ_{r_3} (°)	Uncertainty (°)	Standard deviation / rt (3)
0	0.0	0.0	0.0	± 0.5	
15	15.0	15.5	13.5	± 0.5	
30	30.0	32.0	31.5	± 0.5	
45	46.0	46.0	44.5	± 0.5	
60	59.5	59.5	58.5	± 0.5	
80	80.5	79.5	79.5	± 0.5	

Experiment 2 - Law of Refraction

Determine an experimental procedure that allow you to align the laser with the optical axis and a method of testing/verifying alignment

- Point the laser straight forward at a grid tick mark on the paper. Rotate the laser until it is aligned. Check by moving the paper along the optical axis. The laser is aligned when the bright spot does not move laterally as the paper is moved along the optical axis.

Setup

- Place the laser against to the backface (so that it does nto block th reflected beams) and then place the glass block in front. Rotate the block through an angle of incidence of up to 90 degrees. Measure the angle of the block via a protractor placed against the face of the block. Measure the distance between the first reflected beam and the second reflected beam against the front face of the block.

$\Theta_i (^{\circ})$	Δx	
0 ± 0.5	0 ± 0.1	
10 ± 0.5	0.5 ± 0.1	
21.5 ± 0.5	1.0 ± 0.1	
30.5 ± 0.5	1.5 ± 0.2	
44.5	2.0 ± 0.4	
56.0	2.5 ± 0.4	
61.0	2.7 ± 0.4	
76.5	3.0 ± 0.4	
65.5	2.9 ± 0.4	
48.0	2.2 ± 0.3	

Thickness of the block: 1.876cm, 1.885cm, 1.884cm

Experiment 3

3.1

- Upside down F, real image, may be slightly magnified, very blurry



-
- Parallax - the image is behind the lens

- Move the lens closer to the screen: the image becomes more diminished, less blurry/more focused
-

Experimental Design: Describe the projection that forms. and determine a way to move the screen until you get the sharpest, clearest projection. That is, come up with a procedure that will reliably allow you to locate the (real) image.

- With the lens fixed in place, slowly move the viewing screen either forward or backward relative to the light source until the image is sharp.
- Measure the distance from the lens to the screen by measuring the left side of each of the clamps using the ruler in the optical bench.
- To determine observational uncertainty, measure the worst case real images and half these differences by.

Light source position: 26.1cm

Lens position: 52.6 cm

Lens position (cm)	xi	Closest xi (cm)	Farthest xi (cm)
52.6		89.0	89.9
52.6		87.2	89.3
52.6		88.4	89.2

Virtual images

= Moving the screen creates a diffuse and blurry light patch. The patch changes size but it never focuses in place. The rays never focus because as they exit the lens, they diverge, so will never physically cross in front of the lens. Our brain can see these rays and reconstitute an "imaginary" source of where they came from.

Experimental design:

- Look through the lens and see the floating virtual image. Put a narrow and tall object along the same line of sight and adjust it until there is no parallax shift between the reference pin and the virtual image. Then measure the distance between the reference object and the lens.

	xo	Closest xo (cm)	Farthest xi (cm)

3.2

Determine magnification:

Lens position (cm)	Closest d _i (cm)	Farthest d _i (cm)	Closest h _i (cm)	Farthest h _i
51.0	91.5	92.4	3.6	3.8
55.0	87.3	88.8	2.6	2.7
60.0	87.7	88.7	1.9	2.0
49.0	94.6	97.4	4.5	4.8
65.0	90.3	90.6	1.4	1.5

Light position: 21.6cm

measure F with vernier calipers: 2.230cm, 2.190cm, 2.240cm

Experiment 4 - Determine index of refraction

snells law:

$n_a \sin i = n_g \sin(r)$

Precision 0.5

Theta in	theta out 1	theta out 2
0	0.0	0.0
10	6.5	7.0
20	13.5	14.0
30	20.0	19.5
40	26.0	26.5
50	31.0	32.0
60	36.5	36.0
70	40.0	39.0
80	41.5	41.5

Second test: measure critical angle

Trial 1	Trial 2	Trial 3
43.0pm1	43.0\pm1	43.5 pm1

Laser Optical Disc Set

